

JEE MAIN REVISION MASTER: CURRENT ELECTRICITY 2026 6 QUESTIONS 10% WEIGHTAGE ↓24.98%

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 66 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kirchhoff's Laws, Wheatstone Bridge & Potentiometer	38 PYQs	DC Instruments
Very High	Ohm's Law, Drift Velocity & Resistance Temperature	32 PYQs	Conduction

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Kirchhoff's Laws, Wheatstone Bridge & Potentiometer	$\frac{P}{Q} = \frac{R}{S}$ (Wheatstone), $\frac{E_1}{E_2} = \frac{I_1}{I_2}$, $r = R \left(\frac{I_1}{I_2} - 1 \right)$, $S = \frac{I_g}{I - I_g} R$, $R_{\text{series}} = \frac{V}{I_g} - G$
Ohm's Law, Drift Velocity & Resistance Temperature	$I = n e A v_d$, $v_d = \frac{e E \tau}{m}$, $\rho = \frac{m}{n e^2 \tau}$, $R(T) = R_0(1 + \alpha \Delta T)$, $\vec{J} = \sigma \vec{E}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kirchhoff's Laws, Wheatstone Bridge & Potentiometer (Very High Yield — 38 PYQs)

Core Principle: Kirchhoff's junction and loop rules, balanced bridges, potentiometer EMF comparison, and galvanometer conversion.

■ **Caution / JEE Pitfall:** A potentiometer measures EMF with zero drawn current, making it infinitely superior to a real voltmeter which always draws finite current.

■ **Speed Trick:** To convert galvanometer to ammeter: connect small shunt S in parallel. To voltmeter: connect high resistance R in series.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "In the figure, the values of currents I_1, I_2 and I_3 respectively are..." [Ans: N/A]
- [2026 Shift] Q#1 (MCQ): "Given below are two statements: Statement-I: The resistivity of a conductor is independent of its temperature. Statement-II: The resistivity..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Which of the following circuits is correct for verification of Ohm's law?" [Ans: N/A]

3.2 Ohm's Law, Drift Velocity & Resistance Temperature (Very High Yield — 32 PYQs)

Core Principle: Microscopic model of electrical conduction, electron mobility $\mu = v_d/E$, and thermal variation of resistivity.

■ **Caution / JEE Pitfall:** For conductors, $\alpha > 0$ (resistance increases with temperature). For semiconductors, $\alpha < 0$ (resistance decreases with temperature).

■ **Speed Trick:** When a wire is stretched to n times its initial length while preserving volume, its resistance becomes $n^2 R$.

Representative Source PYQs:

- **[2026 Shift] Q#2 (MCQ):** "The number of electrons moving per second through the filament of a lamp of 60 W operating at 120 V is nearly $(e=1.6 \times 10^{-19})$..." **[Ans: C]**
- **[2026 Shift] Q#4 (MCQ):** "Current flowing through a wire decreases linearly from 10 A to zero in 4 s as shown in the graph. Find the total charge flowing through ..." **[Ans: B]**
- **[2025 Shift] Q#2 (MCQ):** "Given, a current carrying wire of non-uniform cross-section, which of the following is constant throughout the length of wire?..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

In the figure, the values of currents I_1, I_2 and I_3 respectively are

- (A) 6 A, 1.5 A and 1 A
- (B) 4 A, 2.5 A and 2 A
- (C) 4 A, 2.5 A and 1 A
- (D) 6 A, 4.5 A and 1.5 A

Correct Answer: None

Solution Analysis: At any junction, Kirchhoff's Current Law says that the total current entering the junction is equal to the total current leaving it. For the first junction, 5A enters. The currents leaving are 1A and I_1 . So, $5 = 1 + I_1 \Rightarrow I_1 = 5 - 1 = 4A$ For the second junction, I_1 enters. The currents leaving are 1.5A and I_2 . So, $I_1 = 1.5 + I_2$ Putting $I_1 = 4A$, $4 = 1.5 + I_2 \Rightarrow I_2 = 4 - 1.5 = 2.5A$ For the third junction, I_2 enters. The currents leaving are 1.5A and I_3 . So, $I_2 = 1.5 + I_3$ Putting $I_2 = 2.5A$, $2.5 = 1.5 + I_3 \Rightarrow I_3 = 2.5 - 1.5 = 1A$

Benchmark Problem 2: KCET 2026 (Q#2)

The number of electrons moving per second through the filament of a lamp of 60 W operating at 120 V is nearly ($e=1.6 \times 10^{-19}C$) _____.

- (A) 6.2×10^{18}
- (B) 6.2×10^{19}
- (C) 3.1×10^{18} **[Correct]**
- (D) 3.1×10^{19}

Correct Answer: C

Solution Analysis: Power of the lamp is $P=60W$ Voltage is $V=120V$ Using the relation $P=VI$ we get the current through the filament: $I=(P / V)$ $I=(60 / 120)$ $I=0.5A$ Now, current means charge flowing per second. So charge flowing per second is 0.5C Charge of one electron is $e=1.6 \times 10^{-19}C$ Number of electrons flowing per second is $n=(I / e)$ $n=\frac{0.5}{1.6 \times 10^{-19}}$ $n=3.125 \times 10^{18}$ Nearly, $n \approx 3.1 \times 10^{18}$ So the correct option is 3.1×10^{18} Option C.